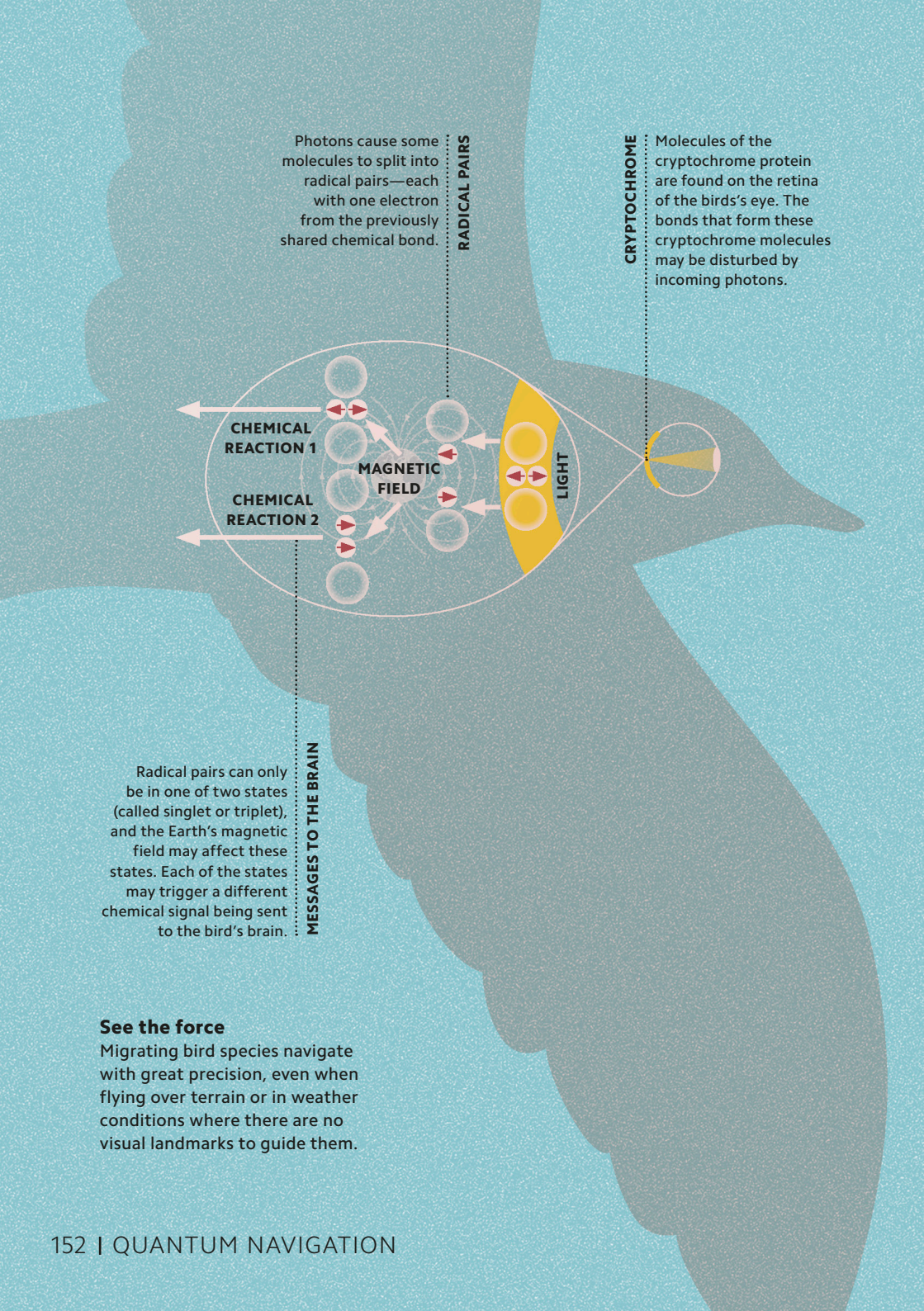




Photons cause some molecules to split into radical pairs—each with one electron from the previously shared chemical bond.

RADICAL PAIRS

CRYPTOCROME

Molecules of the cryptochrome protein are found on the retina of the bird's eye. The bonds that form these cryptochrome molecules may be disturbed by incoming photons.

Radical pairs can only be in one of two states (called singlet or triplet), and the Earth's magnetic field may affect these states. Each of the states may trigger a different chemical signal being sent to the bird's brain.

MESSAGES TO THE BRAIN

See the force

Migrating bird species navigate with great precision, even when flying over terrain or in weather conditions where there are no visual landmarks to guide them.